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METHOD OF STATEMENT FOR AIR CONDITIONING & MECHANICAL VENTILATION (ACMV) (SITE EQUIPMENT PERFORMANCE OF AIR BALANCING TESTING & COMMISSIONING)

Prepared BY: _____
Subcontractor (Name & Signature) (Date)

Checked BY: _____
Project(M&E) (Name & Signature) (Date)
In- Charge

Approval BY: _____
Consultant (Name & Signature) (Date)

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1. Objective

This document outlines the systematic procedure for the testing, commissioning, and air balancing of the Air Conditioning and Mechanical Ventilation (ACMV) system. The primary goal is to verify and document that the ACMV system and its components are installed, operating, and performing in accordance with the project design specifications require. This process ensures the system provides the intended indoor environmental quality, thermal comfort, and energy efficiency.

2. Scope

This method statement applies to the testing and commissioning of the following ACMV systems and equipment:

- Air Handling Units (AHUs) with and without Variable Speed Drives (VSDs) or speed regulators.
- Ducted and wall-mounted Fan Coil Units (FCUs).
- Fresh Air Fans (FAFs) and Exhaust Air Fans (EAFs) for mechanical ventilation, with and without VSDs or speed regulators.
- Associated ductwork, grilles, diffusers, louvers, and dampers.
- Control systems, including the Building Management System (BMS).

The scope of work includes pre-commissioning checks, performance testing, air and water balancing, functional performance testing, and documentation.

3. References

All testing and commissioning activities will be carried out in compliance with the latest editions of the following Singapore Standards and codes of practice:

- **Air-conditioning and mechanical ventilation in buildings:** This standard provides guidance on the design, construction, installation, testing, commissioning, operation, and maintenance of ACMV systems.
- **Indoor air quality for air-conditioned buildings:** This code specifies acceptable indoor air quality parameters to minimize health risks for building occupants.
- **Workplace Safety and Health Act (WSH Act):** To ensure all on-site activities are conducted safely.
- **Building and Construction Authority (BCA) Green Mark Criteria:** To meet the specific requirements for energy efficiency and sustainable building practices.
- **Project-specific design drawings and technical specifications.**

- **Equipment manufacturer's installation, operation, and maintenance manuals.**

4. Responsibilities

- **Project Manager:** Overall responsibility for the successful execution of the project, ensuring all resources are available.
- **Project Coordinator:** To provide technical guidance, review the method statement and test reports. To witness and verify the testing and commissioning process.
- **Commissioning Supervisor:** To plan, coordinate, and execute all testing and commissioning activities. To ensure that all procedures are followed and documented.
- **ACMV Technicians:** To carry out the installation, pre-commissioning checks, and testing and commissioning works under the supervision of the Commissioning Manager.
- **Controls Contractor:** To be present during the testing of control systems and to make any necessary adjustments to the BMS.
- **Safety Personal:** To ensure that all on-site activities comply with the WSH Act and project-specific safety requirements.

5. Preparation

Before the commencement of testing and commissioning, the following preparations shall be completed:

- Submission and approval of this method statement.
- To follow up with approval of all required shop drawings.
- Verification that the ACMV system installation is complete and in accordance with the approved drawings and Project require.
- Ensuring that all necessary utilities, such as power and water, are available.
- Verification that all testing equipment is calibrated and in good working condition.
- Conducting a toolbox meeting with all relevant personnel to discuss the scope of work, procedures, and safety precautions.

6. Site Equipment Performance of Air Balancing Testing & Commissioning

A comprehensive range of calibrated testing equipment will be utilized to ensure accurate measurement and verification of the ACMV system's performance.

7. Pre-Commissioning Checks

Prior to functional testing, the following pre-commissioning checks shall be performed and documented:

- **General:**
 - Verify that all equipment and materials are installed as per the approved submittals and shop drawings.
 - Check for any physical damage to the equipment.
 - Ensure proper access for maintenance and testing.
- **Ductwork:**
 - Visually inspect ductwork for proper sealing and insulation.
 - Confirm that all dampers (volume control, fire, and smoke) are in the correct open/closed position.
- **Fans (AHU, FCU, FAF, EAF):**
 - Check for correct fan rotation.
 - Verify that fan belts are properly tensioned.
 - Ensure that vibration isolators are correctly installed.
- **Piping:**
 - Ensure that all chilled water piping has been flushed, pressure tested, and insulated.
 - Verify that all valves are in their correct positions.
- **Electrical:**
 - Confirm that all electrical wiring is complete and terminated correctly.
 - Verify that all motors have the correct overload protection settings.
 - Check that all electrical panels are energized and safe to operate.
- **Controls:**
 - Verify the installation of all sensors, actuators, and controllers.
 - Check that all control wiring is correctly connected.

8. Testing Equipment & Hand Tools Photos

Category	Item Description
Test Equipment	1. Air Capture Hood 2. Vane Anemometer or HOT wire Anemometer 3. Multi-function Anemometer with various probes (pitot tube, temperature/humidity probe) 4. Multi Clamp Meter
Safety Equipment	1. Safety Barriers. 2. Laminated, Warning Signs. 3. Full-body harnesses and lanyards for any work at height. 4. Ladder, Scaffold, scissor lift if necessary
PPE	1. Safety Helmets. 2. Safety Goggles. 3. Safety Boots. 4. Hand Gloves.

 Air Capture Hood Instruments	 Vane Anemometer	 Multi-function Anemometer
 Multi Clamp Meter	 PPE	 Ladder

9. Using of Equipment with Testing Method

- Air Capture Hood (for 4-way supply/return grilles and linear diffusers):

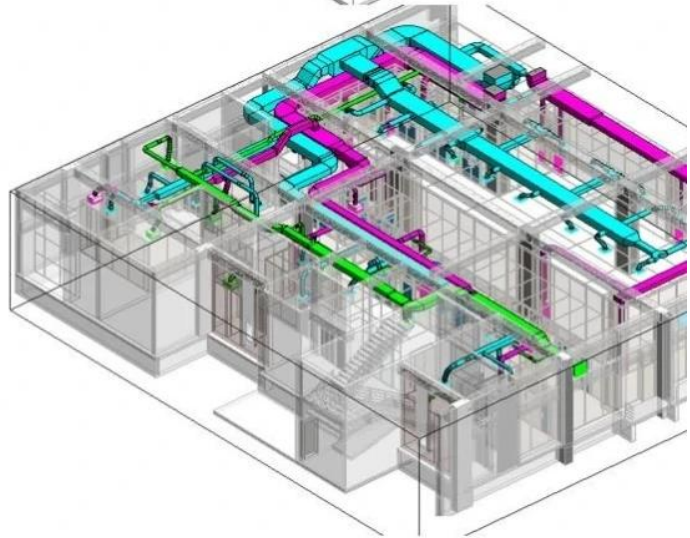
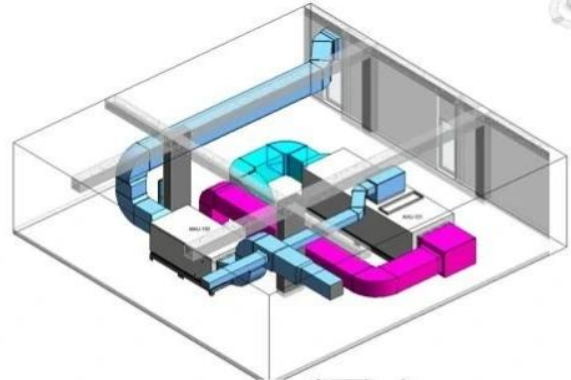
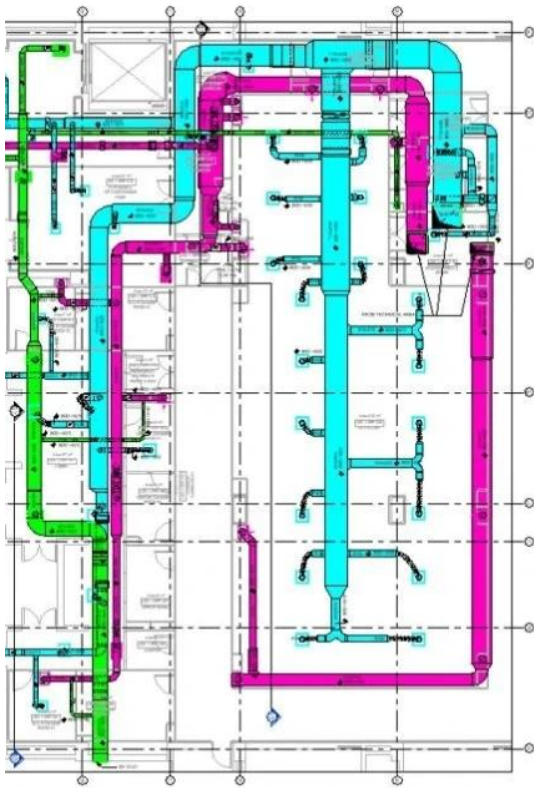
9.1. Using the Air Capture Hood

- **Application:** Preferred method for measuring airflow from individual supply air diffusers (e.g., 4-way, 3-way, 2-way) and return/exhaust air grilles due to its high accuracy.
- **Step-by-Step Procedure:**
 1. **Preparation:** Ensure the air capture hood is assembled correctly with the appropriate skirt size (e.g., 600mm x 600mm, 1200mm x 600mm) that matches the nominal size of the diffuser or grille being tested.
 2. **Power On and Zero:** Turn on the digital meter of the hood and allow it to stabilize. Perform a zero-calibration as per the manufacturer's instructions before taking the first measurement.
 3. **Positioning the Hood:** Place the hood firmly against the ceiling or wall, ensuring the skirt creates a complete seal around the perimeter of the diffuser or grille. This prevents air leakage and ensures all airflow passes through the meter.
 4. **Taking the Reading:** Allow the reading on the digital display to stabilize for several seconds. The meter directly displays the volumetric airflow rate in **Liters per second (L/s)** or **Cubic Meter Per Hour (m³/h)**
 5. **Recording Data:** Record the measured airflow on the "Air Balancing Test Sheet" corresponding to the specific grille/diffuser tag number.
 6. **Comparison and Adjustment:** Compare the measured airflow with the design airflow. If the reading is outside the acceptable tolerance (typically +/- 10% of design), locate the nearest Volume Control Damper (VCD) in the branch ductwork. Adjust the VCD lever and re-measure the airflow until the reading is within the specified tolerance.
 7. **Marking Position:** Once the correct airflow is achieved, mark the final position of the VCD lever with a permanent marker to prevent future maladjustment.
 8. **Repeat:** Repeat this process for all diffusers and grilles in the designated area.



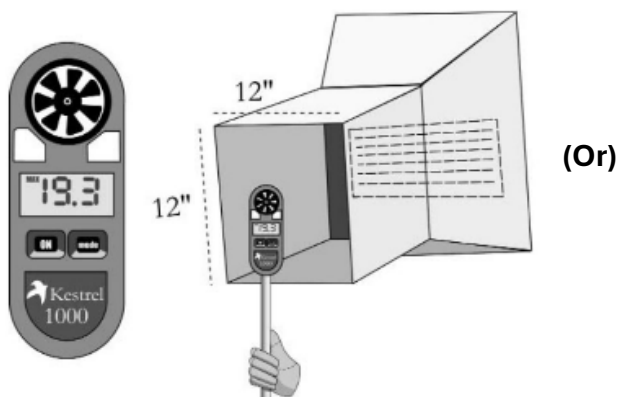
Air Capture Hood

Example Plan Layout Drawings for MV System



10. Using the Vane Anemometer or HOT wire Anemometer

- **Application:** Used for large grilles, louvers, or where an air capture hood cannot be deployed. This method calculates airflow by measuring air velocity over a defined area.
- **Step-by-Step Procedure:**
 1. **Power On and Zero:** Turn on the digital meter of the Anemometer and allow it to stabilize. Perform a zero-calibration as per the manufacturer's instructions before taking the first measurement.
 2. **Measure Grille Area:** Accurately measure the dimensions (length and width) of the grille or louver. Calculate the "free area "based on the Approval drawing. The free area is the actual open area through which air can pass.
 3. **Traverse Pattern:** Mentally divide the face of the grille into a grid of equal, smaller areas (a minimum of 9-12 readings is recommended for accuracy) or Depend on the grille size.
 4. **Positioning the Anemometer:** Hold the vane anemometer head parallel to the grille face at the Center of the first grid section. Ensure the vanes are perpendicular to the direction of airflow.
 5. **Take Velocity Readings:** Record the air velocity reading in meters per second (m/s). Move the anemometer to the Center of each subsequent grid section and record the velocity. This process is known as "traversing" the grille.
 6. **Calculate Average Velocity:** Sum all the recorded velocity readings and divide by the number of readings taken to find the average air velocity.
 7. **Calculate Airflow:** Calculate the volumetric airflow using the formula:
 - **Airflow (m³/hr) = Average Velocity (m/s) × Free Area (m²) × 3600**
 8. **Recording and Adjustment:** Record the calculated airflow on the test sheet. If adjustment is needed, follow steps 6 and 7 from the Air Capture Hood procedure to adjust the VCD and re-measure.



Vane Anemometer

(Or)



Air Velocity Meter

11.0. Duct Static Pressure and Room Condition Measurements

Equipment Required

- Calibrated Multi-function Meter (e.g., TSI VelociCalc 9600 series or equivalent)
- Static Pressure Probe (with side-sensing ports)
- Flexible, color-coded tubing
- Drill with appropriate bit size (if test ports are not pre-installed)
- Duct plugs or high-quality aluminium foil tape
- Safe access equipment (e.g., ladder, mobile scaffold)
- Appropriate Personal Protective Equipment (PPE): safety glasses, gloves, safety footwear.
- Approved Air Balancing Test Sheets for data recording.

11.1. Step-by-Step Measurement Procedure

Phase 1: Pre-Measurement Preparation

1. **Verify System Operating Status:** Confirm with the Commissioning Supervisor that the target AHU/Fan is operating under normal, stable, and full-load conditions. All associated dampers should be in their design position.
2. **Adjust Fan Speed (for VSD units):** Compare the design with the current design static pressure for the AHU or Fan. If there is a discrepancy, adjust the fan speed via the Variable Speed Drive (VSD) control panel or BMS until the design flow is achieved. For non-VSD units, pulley adjustments may be required.
3. **Identify Test Locations:**
 - **Downstream (Outlet):** Locate the designated test port in the main supply/discharge duct, positioned after the fan and any internal components (like cooling coils), but before any main branch take-offs. A straight duct section is preferred.
 - **Upstream (Inlet):** Locate the designated test port in the main return/inlet duct, positioned just before the air enters the AHU/Fan housing (before the filter bank).
4. **Prepare the Multi-function Meter:**
 - Power on the meter and allow the sensor to stabilize.

- Navigate the menu to select the "Pressure" measurement function.
 - Ensure the display units are set to Pascals (Pa).
 - Connect the static pressure probe to the meter's positive (+) pressure port using the flexible tubing.
5. **Zero the Sensor:** With the probe held in ambient air away from any drafts, perform the zeroing function on the meter as per the manufacturer's instructions. This step is critical for accurate readings.

11.2. Phase 2: Execution of Measurement

Measuring Downstream (Positive) Static Pressure:

1. **Access Test Port:** Safely access the downstream test port on the supply duct.
2. **Insert Probe:** Insert the static pressure probe into the duct, ensuring it is perpendicular to the direction of airflow. The tip should be positioned approximately in the center of the duct.
3. **Read and Record:** Allow the reading on the meter to stabilize completely. Record this positive pressure value in Pascals (Pa) on the test sheet under "Downstream Static Pressure" or "Outlet SP".

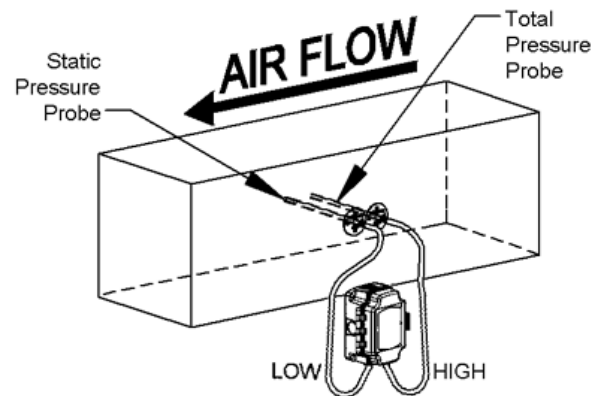
Measuring Upstream (Negative) Static Pressure:

1. **Access Test Port:** Safely access the upstream test port on the return duct.
2. **Insert Probe:** Insert the same probe setup into the return duct, again ensuring it is perpendicular to the airflow.
3. **Read and Record:** Allow the reading to stabilize. The meter will display a negative value (e.g., "-150 Pa"). Record this exact value on the test sheet under "Upstream Static Pressure" or "Inlet SP".

11.3. Phase 3: Calculation and Completion

1. **Calculate Total External Static Pressure (TESP):** The TESP is the total resistance the fan works against. It is not the sum of the absolute values, but the difference between the outlet and inlet pressures.
 - Formula: $\text{TESP (Pa)} = [\text{Downstream SP}] - [\text{Upstream SP}]$
 - Example:
 - Downstream SP = +275 Pa
 - Upstream SP = -160 Pa

- $TESP = (275) - (-160) = 275 + 160 = 435 \text{ Pa}$
- 2. **Record Calculation:** Record the calculated TESP on the test sheet. This value, along with the measured total airflow, will be used to verify the fan's performance against its curve.
- 3. **Seal Test Ports:** After removing the probe, securely seal both test ports using the appropriate plugs or tape to prevent air leakage.
- 4. **Finalize Documentation:** Ensure the test sheet is completed with the date, time, equipment ID, technician's name, and all recorded and calculated values. Submit the completed sheet to the Project Manager for review and submission.



TSI Multi-function Meter

12.0. Room Temperature and Relative Humidity (RH)

1. **Setup Equipment:** Attach the appropriate temperature and RH probe to the multi-function meter.
2. **Verify System Operating Status:** Confirm with the Commissioning Supervisor that the target AHU/Fan is operating under normal, stable, and full-load conditions. All associated dampers should be in their design position.
3. **Position Probe:** Hold the probe in the Center of the room, away from direct sunlight, supply air diffusers, or heat-generating equipment to get a representative ambient reading.
4. **Record Readings:** Allow the readings to stabilize for a few minutes. Record the room temperature (°C) and Relative Humidity (%). Verify these readings are within the comfort.



TSI Multi-function Meter

13. Documentation and Handover

Upon successful completion of all testing and commissioning works, a comprehensive report will be compiled and submitted for approval. This report will include:

- This approved Method Statement.
- Calibration certificates for all testing equipment.
- Completed test sheets for all equipment, including air balancing results.
- A summary of the final performance data against the design intent.
- A list of any outstanding issues and a plan for their resolution.

A formal handover of the commissioned system will be conducted with the client and facilities management team, including a demonstration of the system's operation.

14. Health and Safety

All work will be carried out in strict adherence to the project's Health, Safety, and Environment (HSE) plan and the WSH Act. Key safety measures include:

- A risk assessment shall be conducted prior to the commencement of work.
- Ensure stable and safe access to all test locations. If working at height, appropriate fall protection measures must be in place.
- Verify that the work area is clear of any slip, trip, or fall hazards.
- Technicians must be cautious of moving parts (fan belts, motors) and hot surfaces within the AHU. Do not open access panels while the unit is operating unless specifically required and deemed safe.
- Ensure test ports are properly sealed after measurement to prevent air leakage and potential injury from high-pressure air streams.
- All personnel will wear the appropriate Personal Protective Equipment (PPE).
- Lock-out/Tag-out (LOTO) procedures will be implemented before any work is carried out on energized equipment.
- Safe access will be ensured for all work areas, especially at heights.
- Good housekeeping will be maintained at all times.